Ashwin Chembu

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PROFILE

Data engineer building Snowflake SQL pipelines, healthcare entity-matching workflows, and traceable quality controls. Connects layered ETL/MDM architecture with API-assisted reconciliation and stakeholder-ready evidence.

TECHNICAL SKILLS

Languages: SQL, Python, TypeScript, R

Data Platforms: Snowflake, MongoDB, AWS S3, AWS Glue, Pandas, NumPy

Data Engineering: Layered ETL/MDM, entity resolution, identifier matching, delta logic

Quality / Delivery: Reconciliation, data profiling, source-to-target mappings, QC reports, stewardship review

EXPERIENCE

Business Technology Solutions Associate | ZS Associates — South San Francisco, CA | Jun 2024-Present

- Developed **Snowflake SQL** pipelines across 327K+ records for identifier matching, specialty joins, address comparison, and quality reporting.
- Designed entity-matching frameworks across 67K+ records using healthcare identifiers and fuzzy-search signals for stewardship review.
- Defined layered **ETL/MDM** flows with landing, staging, canonical, and publication layers, including delta logic, history, and quality checks.
- Translated ambiguous stakeholder requirements into testable rules, API integrations, source-to-target mappings, and delivery documentation.

Founder and Lead Developer | Tallyrus — Jun 2023-Present

- Built Flask services with **MongoDB** persistence and S3 document storage to manage application data and support essay-feedback workflows.
- Developed essay-feedback workflows with **Python,** LangChain, Pinecone, and OpenAI to support retrieval-augmented grading for educators.
- Built React/TypeScript and Express flows with Google OAuth, **MongoDB** persistence, secure sessions, and protected application routes.

Data Analyst / Web Development Lead | Tech4Good at UCSC — Mar 2023-Present

- Built Excel/XLSX export workflows and **Python**/Pandas analyses over 43.3K anonymized interaction rows spanning 2.8K verified session IDs.
- Separated state across three **MongoDB** logical databases, using change streams and persisted responses to synchronize edits and votes.
- Implemented 20+ attributed Node/Express REST routes with **MongoDB** persistence and logging for shared chat, text entry, and polling.

PROJECTS

Forge | Personal Fitness Intelligence — Personal project

- Built FastAPI services integrating Hevy, Oura, and Apple Health webhooks into **MongoDB** to unify training, nutrition, and recovery records.
- Implemented six-hour synchronization, on-demand refresh, and CSV ingestion to maintain fitness records across external service integrations.
- Developed an engine comparing performance with a 12-week hypertrophy plan to score workout adherence, nutrition compliance, and recovery.

Azure RAG | Document Search Proof of Concept — Public proof of concept

- Built document search with **Python,** Azure OpenAI embeddings, LangChain, and FAISS similarity search to retrieve relevant source passages.
- Implemented PDF/TXT/Markdown ingestion, text chunking, vector indexing, and source citation to ground interactive Q&A in document context.
- Documented Azure embedding and chat-model setup with a 30+ document corpus for repeatable retrieval and source-grounded Q&A workflows.

Lume | AI Meeting Intelligence — Dec 2025-Present

- Built React and Node/Express workflows with **MongoDB** and S3 to capture, store, search, and query meeting transcripts and project files.
- Implemented reusable Project Brain knowledge bases that retain terminology and relationships from transcripts and seed files for grounded Q&A.
- Designed a RAG engine with vector embeddings, intent routing, evidence extraction, and generator/critic refinement for citation-backed answers.

EDUCATION

University of California, Davis | **B.S. Computer Science** | **GPA: 3.7** — Graduated Jun 2025

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google Data Analytics Certificate

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu